ASSIGNMENT #02: DATA CLEANING AND INTEGRATION WITH PYTHON & SQL

ASSIGNMENT #02: First Data Download & Exploration

Data Integration with AI Skills

ASSIGNMENT OVERVIEW

FIELD	VALUE
ASSIGNMENT NUMBER	2
TITLE	First Data Download & Exploration
GRADE VALUE	10 points
DUE DATE	March 9, 2026, 11:59 PM PT

OBJECTIVES

After completing this assignment, you will be able to:

- Successfully use AI assistants to download real agricultural data using pre-built skills
- Understand how field boundaries connect to crop data via `field_id`
- Observe and understand how datasets are merged automatically by AI
- Create your first interactive web map to explore data patterns
- Gain confidence working with AI assistants for data tasks

INSTRUCTIONS

Step 1: Switch to Your Feature Branch

Ensure you are working in your assignment branch. In your VS Code: terminal (Ctrl+`), run:

```
git checkout feature/assignment-02-first-data
```

Tip: You can ask OpenCode to: *“Create and switch to a new branch called ‘feature/assignment-02-first-data’.”*

Step 2: Prepare the Environment

Ensure your virtual environment is active and the agri-toolkit is ready.

1. Open VS Code (In Codespace or Local) from where you left off on assignment 1

2. Activate OpenCode in the terminal.

Step 3: Download Field Boundaries (AI-Powered)

Your goal is to download field boundaries using the pre-built skill. **You will NOT write code**—the AI assistant will run the example script for you.

Task 3.1: Choose Your Region

Ask OpenCode:

> *"Using the field-boundaries skill, download 20-30 fields from [YOUR_CHOSEN_REGION]. Save to data/assignment-02/fields_EPSG4326.geojson"*

Choose from regions:

- `corn_belt` (Iowa, Illinois, Indiana)
- `great_plains` (Nebraska, Kansas)
- `southeast` (Georgia, Alabama)

Watch as OpenCode:

- Runs the example script from `skills/examples/01_field_boundaries_example.py`
- Downloads the data
- Shows you the results

Document: Which region did you choose? Why?

Step 4: Download CDL Crop Data (AI-Powered)

Next, get crop history for your fields.

Task 4.1: Get Crop Classifications

Ask OpenCode:

> *"Using the cdl-cropland skill, get crop data for my fields for years 2020-2023. Save to data/assignment-02/cdl_EPSG4326.csv"*

Watch as OpenCode:

- Downloads CDL data for your field locations
- Shows what crops were found
- Displays the CSV file contents

Document: What years did you request? What crop types were found?

Step 5: Merge the Datasets (AI-Powered - Watch the Magic!)

This is the key learning moment: **Watch how the AI joins your datasets.**

Task 5.1: Join Fields with Crop Data

Ask OpenCode:

> *"Merge my field boundaries with the CDL crop data using field_id. Save the combined data to data/assignment-02/fields_with_crops.geojson"*

Watch as OpenCode:

- Shows you the Python code it generates
- Performs the merge using `pandas.merge()` on `field_id`
- Creates the joined dataset

Key Concept: The `field_id` column is the "key" that connects field boundaries to crop data. This is how different datasets talk to each other.

Document:

- How many fields have crop data?
- Were there any fields without matching crops?
- What does the merged data look like?

Step 6: Create Interactive Web Map (AI-Powered)

Visualize your data!

Task 6.1: Generate Web Map

Ask OpenCode:

> *"Using the interactive-web-map skill, create a map showing my fields colored by crop type. Save to data/assignment-02/my_fields_map.html"*

Open the map:

- Double-click the HTML file
- Explore: Click on fields, toggle layers, zoom in/out

- Take a **screenshot** for submission

Document:

- What patterns do you see in the crop distribution?
- Any surprises or interesting findings?
- What questions do you have about what you're seeing?

Step 7: Optional Stretch - Add Soil Data

Optional (not required for full credit): If you're curious, try adding soil data.

Ask OpenCode:

> *"Using the ssurgo-soil skill, get soil data for my fields. Save to data/assignment-02/soil_EPSG4326.csv. Then merge it with my fields and update the web map."*

Note: This is bonus exploration—not required for assignment completion.

Step 8: Update Your Project Tracker

Open `docs/project/assignment-02-tracker.md` (or create it).

Mark complete:

- Downloaded field boundaries
- Downloaded CDL crop data
- Merged datasets using field_id
- Created interactive web map

Document any Issues Encountered:

- Did any downloads fail?
- Were there fields without crop matches?
- What questions do you still have?

Step 9: Commit and Push

Ask OpenCode:

> *"Stage all files in data/assignment-02/, commit with the message 'Complete Assignment 4: First data download and exploration', and push to GitHub."*

Step 10: Create Pull Request

Ask OpenCode:

> *"Create a pull request for feature/assignment-02-first-data to main with title 'Assignment 4: First Data Download' and add @borealBytes as reviewer."*

IMPORTANT: Add Instructor as Collaborator

If your repository is **private**, you must add the instructor (@borealBytes) as a collaborator so they can review your work:

To Add a Collaborator:

1. Go to your repository on GitHub
2. Click **Settings** tab
3. Click **Collaborators** (under Access section)
4. Click **Add people**
5. Enter: `borealBytes`
6. Click **Add** and select **Write** permissions

Note: Without this step, the instructor cannot see or grade your private repository.

DELIVERABLES

1. **GitHub Repository Link:** Submit the URL to your repository showing the downloaded data in `data/assignment-02/`
2. **One Screenshot:** Show your web map with fields colored by crop type
3. **Updated Tracker File:** Your markdown tracker reflecting completion of all steps
4. **Pull Request:** Link to your PR with @borealBytes as reviewer

GRADING CRITERIA (10 POINTS TOTAL)

CRITERION	POINTS	DESCRIPTION
Field Boundaries Download	3	Successfully downloaded field data using AI assistant
CDL Data Download	2	Successfully retrieved crop classifications

Data Merge Observed	2	Merged datasets completed, student documented understanding
Web Map Created	2	Interactive map generated and screenshot provided
Git Workflow	1	Proper branch, commits, and pull request created

Note: This assignment is graded on **completion**, not code quality. The goal is successful data download and exploration using AI assistance.

SUPPORT

If you get stuck:

1. **Ask OpenCode:** "I'm stuck on [specific step]. What should I check?"
2. **Check the example scripts:** `packages/agri-data-toolkit/skills/examples/`
3. **Review the presentation slides:** `docs/classes/04-Clean-Fields-Clean-Data-REVISED.md`

Remember: The AI assistant does the coding—you just need to describe what you want!

KEY LEARNING OUTCOMES

By completing this assignment, you will have:

- Downloaded **real agricultural data** (not sample data)
- Seen how **datasets connect** via common keys (field_id)
- Created a **professional interactive map**
- Used **AI assistants** to handle all technical work
- Built confidence for **future data analysis**

You did not write any code. You described what you wanted, and the AI made it happen.